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Java SE 27 & JDK 27

Class CharArrayWriter

java.lang.Object
    java.io.Writer
        java.io.CharArrayWriter

All Implemented Interfaces:

Closeable, Flushable, Appendable, AutoCloseable

```
public class CharArrayWriter
extends Writer
```

This class implements a character buffer that can be used as a Writer. The buffer automatically grows when data is written to the stream. The data can be retrieved using toCharArray() and toString().

Note: Invoking close() on this class has no effect, and methods of this class can be called after the stream has closed without generating an IOException.

Since:

1.1

Field Summary

Fields

Modifier and Type	Field	Description
protected char[]	buf	The buffer where data is stored.
protected int	count	The number of chars in the buffer.

Fields declared in class Writer

lock

Constructor Summary

Constructors

Constructor	Description
CharArrayWriter()	Creates a new CharArrayWriter.
CharArrayWriter(int initialSize)	Creates a new CharArrayWriter with the specified initial size.

Method Summary

All Methods	Instance Methods	

Modifier and Type	Method	Description
CharArrayWriter	**append**(char c)	Appends the specified character to this writer.
CharArrayWriter	**append** (CharSequence csq)	Appends the specified character sequence to this writer.
CharArrayWriter	**append** (CharSequence csq, int start, int end)	Appends a subsequence of the specified character sequence to this writer.
void	**close**()	Close the stream.
void	**flush**()	Flush the stream.
void	**reset**()	Resets the buffer so that you can use it again without throwing away the already allocated buffer.
int	**size**()	Returns the current size of the buffer.
char[]	**toCharArray**()	Returns a copy of the input data.
String	**toString**()	Converts input data to a string.
void	**write**(char[] c, int off, int len)	Writes characters to the buffer.
void	**write**(int c)	Writes a character to the buffer.
void	**write**(String str, int off, int len)	Write a portion of a string to the buffer.
void	**writeTo**(Writer out)	

	Writes the contents of the buffer to another character stream.

Methods declared in class Writer

`nullWriter`, `write`, `write`

Methods declared in class Object

`clone`, `equals`, `finalize`, `getClass`, `hashCode`, `notify`, `notifyAll`, `wait`, `wait`, `wait`

Field Details

buf

`protected char[] buf`

The buffer where data is stored.

count

`protected int count`

The number of chars in the buffer.

Constructor Details

CharArrayWriter

`public CharArrayWriter()`

Creates a new CharArrayWriter.

CharArrayWriter

`public CharArrayWriter(int initialSize)`

Creates a new CharArrayWriter with the specified initial size.

Parameters:

`initialSize` - an int specifying the initial buffer size.

Throws:

`IllegalArgumentException` - if initialSize is negative

Method Details

write

```
public void write(int c)
```

Writes a character to the buffer.

Overrides:

`write` in class `Writer`

Parameters:

`c` - int specifying a character to be written

write

```
public void write(char[] c,
                  int off,
                  int len)
```

Writes characters to the buffer.

Specified by:

`write` in class `Writer`

Parameters:

`c` - the data to be written

`off` - the start offset in the data

`len` - the number of chars that are written

Throws:

`IndexOutOfBoundsException` - If `off` is negative, or `len` is negative, or `off + len` is negative or greater than the length of the given array

write

```
public void write(String str,
                  int off,
                  int len)
```

Write a portion of a string to the buffer.

Overrides:

`write` in class `Writer`

Parameters:

`str` - String to be written from

`off` - Offset from which to start reading characters

`len` - Number of characters to be written

Throws:

`IndexOutOfBoundsException` - If `off` is negative, or `len` is negative, or `off + len` is negative or greater than the length of the given string

writeTo

```
public void writeTo(Writer out)
             throws IOException
```

Writes the contents of the buffer to another character stream.

Parameters:

`out` - the output stream to write to

Throws:

`IOException` - If an I/O error occurs.

append

```
public CharArrayWriter append(CharSequence csq)
```

Appends the specified character sequence to this writer.

An invocation of this method of the form `out.append(csq)` when `csq` is not `null`, behaves in exactly the same way as the invocation

```
    out.write(csq.toString())
```

Depending on the specification of `toString` for the character sequence `csq`, the entire sequence may not be appended. For instance, invoking the `toString` method of a character buffer will return a subsequence whose content depends upon the buffer's position and limit.

Specified by:

`append` in interface `Appendable`

Overrides:

`append` in class `Writer`

Parameters:

`csq` - The character sequence to append. If `csq` is `null`, then the four characters `"null"` are appended to this writer.

Returns:

This writer

Since:

1.5

append

```
public CharArrayWriter append(CharSequence csq,
                              int start,
                              int end)
```

Appends a subsequence of the specified character sequence to this writer.

An invocation of this method of the form `out.append(csq, start, end)` when `csq` is not `null`, behaves in exactly the same way as the invocation

```
    out.write(csq.subSequence(start, end).toString())
```

Specified by:

`append` in interface `Appendable`

Overrides:

`append` in class `Writer`

Parameters:

`csq` - The character sequence from which a subsequence will be appended. If `csq` is `null`, then characters will be appended as if `csq` contained the four characters `"null"`.

`start` - The index of the first character in the subsequence

`end` - The index of the character following the last character in the subsequence

Returns:

This writer

Throws:

`IndexOutOfBoundsException` - If `start` or `end` are negative, `start` is greater than `end`, or `end` is greater than `csq.length()`

Since:

1.5

append

```
public CharArrayWriter append(char c)
```

Appends the specified character to this writer.

An invocation of this method of the form `out.append(c)` behaves in exactly the same way as the invocation

```
    out.write(c)
```

Specified by:

append in interface Appendable

Overrides:

append in class Writer

Parameters:

c - The 16-bit character to append

Returns:

This writer

Since:

1.5

reset

```
public void reset()
```

Resets the buffer so that you can use it again without throwing away the already allocated buffer.

toCharArray

```
public char[] toCharArray()
```

Returns a copy of the input data.

Returns:

an array of chars copied from the input data.

size

```
public int size()
```

Returns the current size of the buffer.

Returns:

an int representing the current size of the buffer.

toString

```
public String toString()
```

Converts input data to a string.

Overrides:

toString in class Object

Returns:

the string.

flush

```
public void flush()
```

Flush the stream.

The `flush` method of `CharArrayWriter` does nothing.

Specified by:

`flush` in interface `Flushable`

Specified by:

`flush` in class `Writer`

close

```
public void close()
```

Close the stream. This method does not release the buffer, since its contents might still be required. Note: Invoking this method in this class will have no effect.

Specified by:

`close` in interface `AutoCloseable`

Specified by:

`close` in interface `Closeable`

Specified by:

`close` in class `Writer`

Report a bug or suggest an enhancement
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